2023 Homework 3
Verified three-bit static CMOS ripple-carry adder

EE3032 Introduction to VLSI Design

13 October 2026

Requirements and interpretation

The assignment asks for a three-bit ripple-carry adder, arbitrary transistor sizes, a 1 fF capacitance at every adder output, HSPICE verification, worst-case propagation delay and average power. Although the sentence says "four full adders," Fig. 1 shows FA0, FA1 and FA2, with three sum bits. The implementation follows that explicit three-stage diagram. The fourth output is carry-out; it is not a fourth full-adder stage.

Full-adder derivation and CMOS topology

At each stage let $a = x_i$, $b = y_i$, $c = c_i$. The required equations are

$$c_{i+1} = ab + c(a + b), \qquad s_i = a \oplus b \oplus c.$$

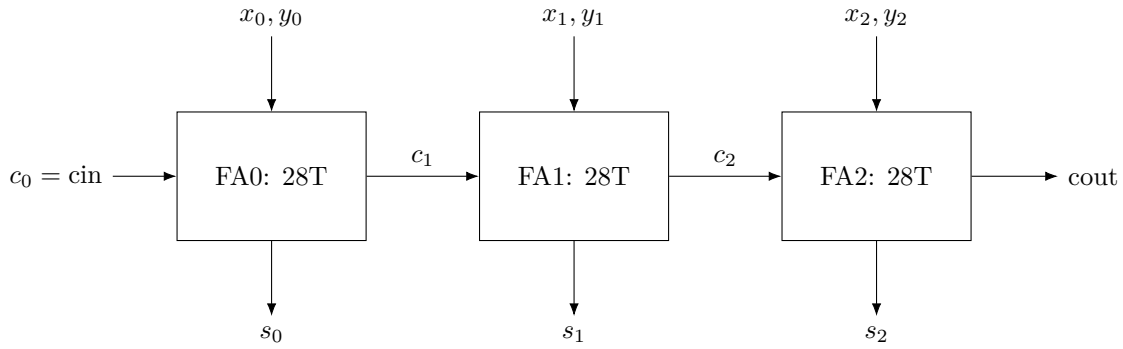
A compact complementary static CMOS implementation uses the internal complemented carry $k = \overline{ab + c(a + b)}$. The sum condition is equivalently $s_i = k(a + b + c) + abc$: with one input high $k = 1$; with two high $k = 0$ and $abc = 0$; with all three high $abc = 1$. Thus the two inverting networks are

$$k = \overline{ab + c(a + b)}, \qquad q = \overline{k(a + b + c) + abc}, \qquad c_{i+1} = \bar{k}, \quad s_i = \bar{q}.$$

The carry gate has ten MOSFETs, the sum gate fourteen, and the two output inverters four: **28 transistors per full adder, 84 total**. Each pull-up is the exact series/parallel dual of its pull-down, so these are complementary static CMOS gates with restored outputs, not floating pass-only signals. All pMOS bodies are at VDD; all nMOS bodies at ground.

Gate	NMOS pull-down (on condition)	PMOS pull-up connectivity
Carry k	a–b in parallel with c–($a \parallel b$)	($a \parallel b$) in series with [$c \parallel (a$–$b)$]
Sum q	k–($a \parallel b \parallel c$) in parallel with a–b–c	($a \parallel b \parallel c$) in series with [$k \parallel (a$–$b$–$c)$]

In the PMOS column labels denote gate signals; a device turns on for a zero gate signal. The exact transistor connections are in `rca.sp`. The independent switch-connectivity audit and electrical simulation both verify the derived arithmetic.



The top-level port order is `cin x0 x1 x2 y0 y1 y2 s0 s1 s2 cout vdd vss`. The binary numerical output is $s_0 + 2s_1 + 4s_2 + 8\text{cout}$.

Sizes and electrical conditions

Device group	Count/FA	W_p (μm)	W_n (μm)
Carry gate	5p + 5n	4	2
Sum gate	7p + 7n	6	3
Two output inverters	2p + 2n	2	1

All channel lengths are $0.18\,\mu$m. These are explicit arbitrary sizes, as permitted by the assignment, selected for complete static CMOS operation. The width-length sum is $3[5(4+2)+7(6+3)+2(2+1)](0.18) = 53.46\,\mu\text{m}^2$. This is a gate-area proxy, not layout area or a claim of optimal sizing. The process is the user's local CIC018 TT BSIM3 library, 1.8 V, 25°C. Each of $s_0, s_1, s_2, \text{cout}$ has exactly one explicit 1 fF load. Internal carry nodes retain their natural next-stage MOS gate load; no extra output capacitors are added to internal carries. Input slew is 20 ps.

Exhaustive functional verification

The state index encodes $x = \text{state} \bmod 8$, $y = \lfloor \text{state}/8 \rfloor \bmod 8$, $\text{cin} = \lfloor \text{state}/64 \rfloor$. The independent expected output is the integer $x+y+\text{cin}$. All **128 input assignments** pass an actual transistor-level transient test, including carry-in zero and one. Each state lasts 3 ns and is sampled 2.9 ns after its start. Accepted bands are $[-0.02, 0.20]$ V for a zero and $[1.60, 1.82]$ V for a one. The result table saves all four measured voltages, expected and observed integers, and pass flags.

Separately, the delay sweep tests all **896 directed single-input transitions**: 128 initial assignments times seven input bits. Every sweep starts from its own DC operating solution, holds the initial input for 1 ns, changes one input during 1.00–1.02 ns, and holds the final state until 3 ns. Before/after samples verify the endpoints for every transition. Unlike a continuously concatenated timing sequence, each case has the same settled initial-condition protocol.

Propagation delay

For every output whose final Boolean value changes, delay is the last 0.9 V output crossing minus the input's 0.9 V crossing at 1.01 ns. The reference source has the same midpoint for rising and falling inputs. Using the last crossing counts any intermediate hazards before final settling. Unchanged outputs have no required functional propagation delay; their failed crossing measurements are intentionally excluded, while their endpoint voltages are still checked. All 1,552 required output transitions have finite positive measured delays and valid endpoint levels.

The maximum is the greater of the measured rise/fall delays, not the arithmetic mean $(t_{pLH}+t_{pHL})/2$. Results use a maximum 2 ps timestep.

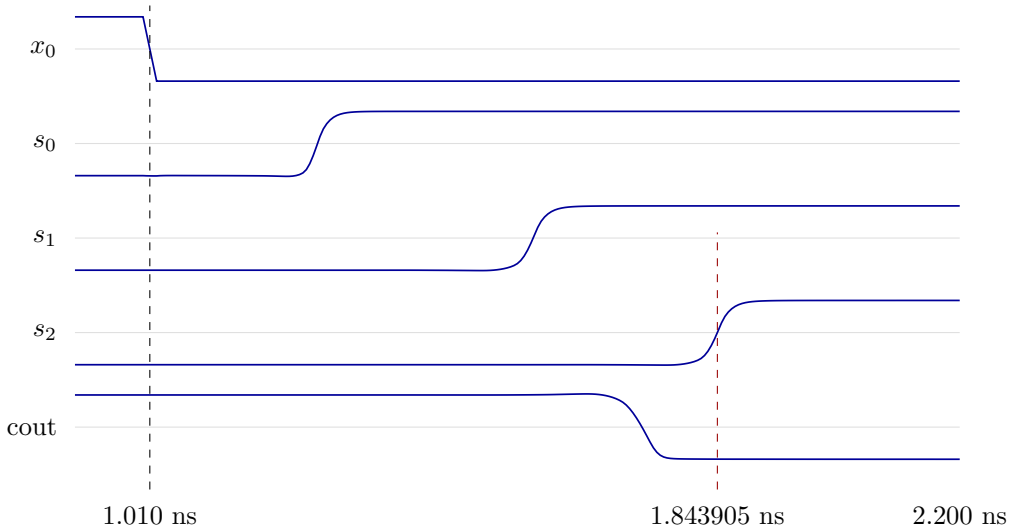
Metric	Measured delay (ps)
Maximum low-to-high	833.526988
Maximum high-to-low	722.903157
Overall maximum	833.526988

The critical case is sweep 498: $x = 7 \to 6$, $y = 0$, cin $= 1$, input `x0` falling, output `s2` rising. The complete output integer changes from 8 to 7, sensitizing ripple-carry removal into the most significant sum bit.

This is an exhaustive worst case over the stated single-input, settled-state characterization at TT, 1.8 V, 25°C, 1 fF and 20 ps input slew. It is not a bound over all simultaneous input changes, glitches, input histories or process/voltage/temperature corners.

Refined critical-transition waveform

The original worst transition was independently rerun with a 1 ps maximum timestep on the same hashed netlist. The refined s_2 delay is 833.905310 ps, versus 833.526988 ps in the complete 2 ps sweep: a 0.378322 ps difference (0.045388%). All four final levels pass. This refines that one transition; it is not a second complete sweep. The traces below use actual saved HSPICE samples, decimated for display.



Each trace spans 0–1.8 V; the dashed lines indicate the input and s_2 output 50% crossings. Full-resolution 3,001-sample data, a PNG plot and refinement metadata are in the results directory.

Average power

The average power workload is explicitly the ordered states 0–127, 3 ns per state, giving a 384 ns interval. All simultaneous binary-count input changes use the same 20 ps slew. VDD supply power is $p(t) = -V(\text{vdd})I(VDD)$ using HSPICE's current sign convention. The testbench averages and integrates this expression over the same interval.

$$\boxed{P_{\text{avg}} = 177.082000\,\mu\text{W}}, \qquad \boxed{E_{384\,\text{ns}} = 67999.487960\,\text{fJ}}.$$

The unit relation is $E[\text{fJ}] = P[\mu\text{W}]\,T[\text{ns}]$; the independently measured integrated energy agrees with $P_{\text{avg}} \times 384$ ns.

Supply power includes dissipation and changes of energy in the circuit's capacitors supplied by VDD, but excludes energy delivered by the ideal input voltage sources. HSPICE's built-in `POWER` uses a different all-independent-sources convention; these numbers are explicitly VDD-only. Power depends on workload and input frequency and is not a universal circuit constant. It is measured in a separate run from the DC-reinitialized timing sweeps, whose initialization energy would not form a realistic workload.

Existing-answer review and integration

Three supplied HW3 reports and their SPICE files were read first: Lin (110501558), Liao (110501512), and Jian (109501570). Lin provides the compact 28T carry/sum architecture. Its HW3 code has several bulk terminals tied to changing internal nodes and explicitly loads only carry outputs. The independent implementation retains the valid Boolean architecture, connects bodies to the rails, chooses explicit sizes, and loads all four external RCA outputs. The supplied Lin HW4 pre-layout code already uses rail-body ties; this supports that correction. Liao uses two 12T XOR gates and three 4T NAND gates, 36T per FA. That architecture is valid, but its reported "worst" delay is an average of one chosen rising/falling carry path rather than an exhaustive maximum. Its NAND cell also ties one NMOS body internally. Jian decomposes the equations into many AND/OR gates and uses 1.2 V stimuli; its selected patterns do not constitute a general worst-case proof.

The selected 28T architecture reduces transistor count relative to the 36T gate decomposition. No lowest-power or minimum-delay claim is made across the differently sized supplied circuits. The combined answer improves body connections, required loading, full input coverage, both delay directions, measurement definitions and reproducibility. Reference numeric values are not relabelled as measurements of this implementation.

Files and reproduction

File	Purpose
`rca.sp`	Complete hierarchical 84T transistor circuit.
`characterize.sp`	896-case DC-initialized timing data sweep.
`testbench.sp`	128-vector workload, output loads, power and samples.
`reproduce.py`	Runs real HSPICE, checks samples, exports result tables.
`results/`	Summary, 128 state results, 896 transition endpoint rows and 1,552 delay rows.

Run `python reproduce.py -work PATH -results results` from the homework folder. Override `-model PATH` and `-hspice PATH` as needed. The proprietary process library remains external. Raw simulator listings and logs remain in the local work directory; compact measured data and runnable source are published with the report.